SafeGuard SA: Problem, Market Check and Approach v1.0

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🔥 Brutal summary

The exact product does not exist in South Africa, but most of the features on your poster already do. CityMenderSA already does hazard reporting, reference numbers, community "still there or fixed" updates and escalation to municipalities. Pitch those features and any judge who knows CityMender will see a copy.

Your defensible idea is narrower and stronger. A reported hazard has two separate outcomes: **made safe** (cones, barricade, temporary cover) and **fixed**. Nobody in SA tracks or shows the first one. The UK has worked this way for years: make safe within 24 hours, permanent repair within 28 days. You are bringing a proven standard to Gauteng, not claiming a world first, and that is the better pitch.

Cut restaurants, shops and hospitals. Lead with the protection clock. Make schools your first users, not municipalities.

Does it already exist?

No single tool does what SafeGuard SA proposes, but reporting and community follow-up are already covered. The gap that stays open is a separate, public record of whether a hazard has been **made safe**, signed off by someone authorised.

Tool	What it already does	What it does not do
CityMenderSA (Durban, iOS, Android and web)	Photo and GPS reports, reference numbers, auto ward routing, escalation by WhatsApp and email, Waze-style community updates, Drive Mode hazard warnings, ward insights, councillor portals, municipal reference lookup. About 2,700 faults logged in its first seven months (Daily Maverick)	No barricade or temporary protection record. No made-safe status. Fixes are "verified" by community updates, satellite and AI, not by a named authority

Tool	What it already does	What it does not do
FixLocal (web and WhatsApp, May 2025)	Tells you who to report to and how to escalate: councillor contacts, media, WhatsApp groups, protest	No case tracking or verification. Nothing on interim safety
JRA Find&Fix and CoJ e-services	Log a fault, get a reference number	Ticket goes open to closed. In 2015 a user said a missing manhole cover sat for four months after being logged twice (Fin24)
Johannesburg Water smart covers (Engineering News, Apr 2025)	Sensor covers at strategic locations, monitored 24/7 for tampering	Hardware at selected sites only. No public record. Does not cover sinkholes or other asset owners
FixMyStreet Pro (UK)	Reports wired into council asset systems. Statuses like Action scheduled, In progress, Fixed	No public "made safe" status
UK highway inspection codes (e.g. Hartlepool 2018)	Dangerous defects made safe or corrected within 24 hours, permanent repair within 28 days, temporary measures re-inspected until fixed	Internal council practice, not a citizen product. No sign of it in Gauteng
SafetyCulture	Checklists, audits and corrective actions for businesses	Private facility tool with no public entry point

What this means: the made-safe layer is a feature gap, not a moat. CityMender could add a "made safe" status in a week. Your edge has to be the workflow (report, make safe, verify, fix, verify) plus the users you own. A button alone won't hold.

Two checks before you pitch: find out what **Civic Eye** (3rd at the June 2026 provincial finale) built, and do a CIPC name search, because "Safeguard" is already used by several security and cyber brands.

The problem

When someone reports an open manhole or sinkhole in Gauteng, the system logs a repair ticket, not a safety status. Nobody can see whether the hole has been made safe while it waits: not the resident, not the school next door, not the councillor, and often not the municipality itself. Repairs wait months for budget and contractors, and in that gap people fall in.

The root cause is a mismatch. Making a hazard safe is cheap and quick: cones, a barricade, a temporary plate. Repairing it is slow and depends on budget. Every system in SA only tracks the repair, so the cheap protective step has no owner, no deadline and no record.

Why it is a problem

When	Where	What happened
Jul 2026	Khutsong Ext 3, by Hlanganani Primary	A collapsing sewer manhole turned into a sinkhole. Residents say they reported it about three months earlier. The road was only cordoned off on 27 July, after a car nearly got stuck. Residents say the cordon is inadequate and children play there. A R6m repair at the same spot had been finished less than a year earlier (Carletonville Herald)
Oct 2025	Khutsong	Two-year-old Ntseka Nkatshumane drowned in a water-filled manhole. Residents say they had reported open manholes for months, after nearly R2m was spent on a manhole-covering project (eNCA)
Apr 2025	Soweto, Chris Hani Rd	*Yende v CoJ*: a pedestrian fell into an uncovered stormwater manhole. JRA conceded its duty to cover or barricade. It inspects lids twice a year, relies on public reports, and was down from six inspectors to four. It had no documents to back its defence, and the court held it 50% liable (SAFLII)
Mar 2025	Khutsong	A sinkhole next to Hlanganani Primary had no fencing, so children could get to it (IOL)

When	Where	What happened
Jun 2022	Soweto	A six-year-old fell into a sewer manhole while playing football near home (Briefly)

Even on CityMender, the average time to resolve a fault across municipalities is 101 days. That is roughly three months of exposure per hazard unless something protects people in the meantime.

One correction to your own documents: your 3-person plan cites an 8 Oct 2026 Herald report on Hlanganani. I could only find the 30 July 2026 article. Confirm the October link before it goes on a slide.

Attempts to solve it so far

Every attempt so far treats "fixed" as the only outcome that counts. None of them owns the weeks between the report and the fix.

Attempt	Examples	Why it falls short
Municipal reporting channels	JRA Find&Fix, CoJ e-services, call centres	Tracks the repair only. The resident gets a reference number, then silence
Civic reporting apps	CityMenderSA, FixLocal	Easier reporting, public pressure, escalation. The status is still broken or fixed, with nothing in between
Spending on covers and repairs	Merafong's R2m manhole-covering project, the R6m Hlanganani repair, Joburg Water's smart covers	Covers get stolen again and repairs fail. A project budget doesn't tell anyone whether a particular hole is protected today
Courts	*Yende v CoJ*	Pays out after someone is hurt, so it doesn't prevent anything. It does show that a municipality with no records loses
Community pressure	WhatsApp groups, councillors, media, protest	Gets attention, but leaves no evidence trail anyone can act on or defend

Value proposition

"Other apps tell you a fault was logged. SafeGuard SA shows whether people are protected while it waits to be fixed, and how long they have gone unprotected."

The app doesn't install barriers. It turns the cheap protective step into something that is visible, owned and on a clock.

User	Their pain today	What SafeGuard SA gives them
School (principal, SGB, safety committee)	A known hazard near the gate, a duty of care to learners, and nothing to escalate with except phone calls	One case file: protection clock, safe-distance photos and every follow-up, ready to send to the councillor, district office or media
Resident or parent	Reports it and hears nothing	Plain-language public status: "Reported 9 days ago. Not made safe."
Ward councillor	Angry residents and no overview	A ward list of uncontained hazards, sorted by days unprotected and distance to schools
Municipal operations (JRA, Joburg Water, local technical services)	Backlog, no budget to fix everything, legal exposure	A make-safe queue with the cheap actions first, plus a timestamped record that they acted. That record is what JRA was missing in *Yende*

The product has to work in two modes. **Without** the municipality on board, its value is the 3evidence record and community pressure. **With** the municipality on board, its value is the make-safe queue and liability protection. Your pitch should show both, because judges will ask "what if government ignores it?"

Who the users should be

Your documents list six user types. That is too many for a hackathon and too many for a first pilot. Start with schools, because they sit at a fixed spot, are organised, carry the highest stakes and have a reason to keep records up to date.

Order	User	Role in the product	Why this order
1	**Schools**: principal, SGB or safety committee	Own the hazards inside a radius of the school. Open cases and keep them updated	Fixed location, organised, duty of care, and the strongest story (Khutsong). One school is one pilot
2	Parents and residents	Report from a safe distance and add "still open" sightings through a WhatsApp link	Reach without asking anyone to install an app. They feed the clock
3	Ward councillor or ward committee	Gets the ward list and escalates	The real escalation path, and politically accountable
4	Responder and authorised verifier	Records make-safe and repair actions. A separate person signs off	Simulated at the hackathon. In a pilot: one willing department, or a private asset owner (estate, campus, business park) that controls its own manholes. Easier to win than a metro
5	Municipality or province	Buys the operations dashboard	Last. Procurement is slow, and you need pilot data first

Do not make learners reporters. They are minors, it would put them near the hazard, and it creates POPIA problems around children's data.

SA constraints to design for: WhatsApp intake and a light web page rather than an app install. Capture should work offline and send later. Support Setswana, Sesotho and isiZulu alongside English. Use a pin plus a landmark ("opposite the school gate"), because the GPS of someone standing at a safe distance marks where they are, not where the hole is.

Best approach

Build one hazard near one school, end to end, with the protection clock as the hero. Your 3-person plan is the right document to follow. Use the PDF only for its safeguards section.

The demo story (five beats, about 90 seconds)

1. A parent reports an open manhole near a school through a WhatsApp-style link. The case opens and the clock starts: "Reported. Not made safe."
2. Two days of community sightings land on the case ("still open, no barrier"). The clock keeps running, and each sighting is labelled community-reported.
3. The councillor's ward view puts it at the top: 9 days unprotected, 400 m from a school.
4. A field team records cones and a barrier. The status reads "Barrier recorded, awaiting verification." Then a separate verifier signs off and the clock stops. The repair track stays open.
5. **The kill shot:** a resident later reports "barrier gone". The case moves back to reassessment, and the clock restarts. Covers get stolen in Joburg and temporary barriers can vanish just as easily, which is why sign-off has to be repeatable rather than a one-time tick.

Screen and data rules

- Two separate tracks on every case: **Made safe** (none, recorded, verified, reassess) and **Fixed** (none, recorded, verified). Never a single status field.
- The clock counts from the first report on SafeGuard SA, and says so. It doesn't guess how old the hazard is.
- Residents can never mark anything safe. Only the verifier role can, enforced on the server, not just by hiding a button.
- Show the official municipal reference number when someone enters one. Never invent one.

If the judges score AI use

The June 2026 round asked teams to use AI to build solutions (Krugersdorp News), and your docs say "no AI". Confirm with an organiser today. If AI counts, add **one** bounded use, labelled as a suggestion: flag a likely duplicate within 50 m, blur faces and number plates in photos, or turn a voice note into a written report. Never AI severity scores, and never AI clearing a hazard.

After the hackathon: a 30-day pilot with one school

Pick one school in Khutsong or Soweto, with the SGB's agreement. Log every hazard within 1 km for 30 days and track five things: hazards logged, median days unprotected, the share with an official reference number, the share made safe within 7 days, and how many the councillor escalated. That data is your pitch to a municipality. Without it you are selling an opinion.

Mentor review

🧱 What's broken or missing

- **Three documents describe three different products.** The poster is a community reporting app. The concept proposal is a municipal workflow for 10 people. The PDF is a multi-sector safety platform that includes restaurants and hospitals. Judges need one product.
- **The poster leads with what CityMender already has.** Hazard reporting, community follow-up and official reference tracking take up three of your seven features. The protection clock, your real idea, sits second.
- **Nobody has been identified as the verifier.** "Qualified verification" assumes a municipal inspector who isn't signed up. There's no plan for who signs off in a pilot.
- **Too many users and no first user.** You list six user types, and none of them is clearly the user you start with.
- **There's an AI risk.** The last round asked for AI, and your docs rule it out completely.
- **Rules aren't confirmed.** Your docs plan for 3 people and for 10. This round's published rules say ten participants per region go to the November finale, and entrants must be 18 to 35, live in Gauteng and register in their own region (htxt).

✅ What's solid

- The two-track state model (made safe vs fixed), the append-only timeline, and separate observed-at and submitted-at times. This is careful thinking, and most hackathon teams won't have it.
- The safety rules: never send residents near the hazard, residents can never mark anything safe, an emergency disclaimer, POPIA minimisation.
- The honesty: no world-first claim and no fake integrations. Judges trust teams that know their limits.
- A local, sourced story (Khutsong, a school, a child).
- The ship gates in the 3-person plan.

🛠️ Fixes

- Rewrite the poster's Key Features to lead with: Protection clock, Made safe vs Fixed, Authorised sign-off, Reopens if the barrier disappears. Move reporting and community follow-up into "How it works".
- Add the "barrier gone, case reopens" beat to the demo.
- Add one slide titled "Doesn't CityMender do this?" using the comparison table above.
- Delete every mention of restaurants, retail, hospitals, EHPs and OHSC from the deck.

📌 Priority plan

When	Do this
Now (today)	Confirm the AI rule and team rules with an organiser. Freeze scope to the 3-person plan. Build the two-track case, the clock and the reopen. Rehearse twice
Next (before the November finale, if selected)	Interview one school principal or SGB member. Call JRA, Joburg Water or Merafong technical services and ask how barricades are logged today (your hypothesis H2). Add WhatsApp intake. Mock up the councillor ward view
Later	Run the 30-day school pilot. Approach one private asset owner as a first paying customer. Talk to CityMender about partnering rather than only competing

🚪 Ship gate

- [ ] The pitch names CityMender and states the difference in one sentence
- [ ] Nothing about restaurants, retail or hospitals appears anywhere
- [ ] Every case shows two tracks, never a single "safe" status
- [ ] A resident account cannot verify, tested on the deployed URL
- [ ] The clock stops only on a verified make-safe and restarts on "barrier gone"
- [ ] Every fact on a slide links to a page you have actually opened
- [ ] The AI rule has been confirmed with an organiser
- [ ] The demo runs twice on a phone over mobile data, with a backup video recorded

Sources

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